

Chad Bartel

Backend Developer – Python, FastAPI, PyTest, AWS

[LinkedIn](#) | [GitHub](#) | [Website](#)

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I am a **Backend Python Developer** specializing in Python, FastAPI, PyTest, Lambda, DynamoDB, S3, Git, and Jira. My current interests include Ansible and homelabbing with Raspberry Pi. I have 7+ years of experience in the industry and I have built applications for the financial and manufacturing industries.

Experience

PENNYMAC – Senior Software Engineer (Python, AWS CDK, Jinja2, FastAPI, Lambda, Kinesis)

January 2025 - Current

Architected and implemented new master data management (MDM) customer preference system to new SaaS process.

- ★ Unified API for Profile event routing (Python, FastAPI, Pydantic, AWS CDK, API Gateway)
- ★ Profile event processor (Python, AWS CDK, Lambda, DynamoDB, Kinesis)
- ★ MDM common code package (Python, Poetry, Nox, Setuptools, GitLab CI, Artifactory)

PENNYMAC – Software Engineer (Python, AWS CDK, Jinja2, FastAPI, Lambda, SQS, New Relic)

July 2021 - December 2024

Refined my backend development by deploying various integrations and automations ranging from internal APIs using Auth0 to complex Amazon Step Functions and event-driven microservices using DynamoDB and S3.

- ★ Developed event-driven microservice Lambda functions with EventBridge and SQS to automate underwriting, check borrower credit, request appraisals, and preparing closing documents
- ★ Leveraged New Relic to expose additional metrics required for automated alerts and tracking performance
- ★ Integrated with third-party APIs while securely handling credentials via Secrets Manager and DynamoDB

Projects

Arcane Scribe ([Code](#) | [View Site](#))

May 2025 – July 2025

A tabletop RPG RAG system served by a Javascript web application and API Gateway using the FastAPI framework. With this, I can upload various PDF rulebooks for tabletop games to help me learn and build characters.

- ★ Built using FastAPI and Pydantic for data validation
- ★ API Gateway with Lambda backend for processing requests
- ★ Cognito for user-management to let my friends use it :)
- ★ 90%+ testing coverage with PyTest using Nox

Interests

Homelab: I have a handful of Raspberry Pis deployed with Portainer (Docker Swarm), Pi-hole, and OpenVPN.

Music Production: I make electronic music with Ableton and my Eurorack synthesizer!

Tabletop Gaming: Started with Dungeons & Dragons 3.5e in middle school and I've been playing new ones recently like Dungeon World, Fate, and Savage Worlds!

Education

Western Washington University

B.S. Mathematics, Bellingham, WA, September 2007 - June 2012